

5ARF0 Case Descriptions

Dear Students in 5ARF0,

In what follows you find a set of case descriptions that serve as topics for your project in this course. “Clients” representing the topic domain have agreed to meet with you at least once during the course to answer your questions about a challenge they face. Your job is to design an AI-based tool, create a mock-up or demo of the tool, and design an interface (a context of interaction) for using the tool. Your design should respond to the challenge identified by the client.

The overview of topics is:

ANWE: AI for Animal Welfare

EADS: AI for Eating Disorder Symptoms

ENSU: AI for Environmental Sustainability

Read the overview of the cases below. After that, your first step is to do a literature review of the target context, helping to identify key issues and challenges that arise in that context. You will use this to motivate your first ideas for a design. The first assignment is an initial proposal. You find the requirements for that proposal on Canvas.

Good luck and best wishes,

The Instructor

Topic ANWE

Artificial intelligence (AI) could dramatically impact the lives of non-human animals, opening many possibilities to exploit and control them, but also to understand and communicate with them in a hitherto unimaginable way. The impacts remain underexplored, and ethical guidelines on AI rarely discuss animals (Owe and Baum 2021; Ryan 2023). How can we ensure that AI-driven applications in this field are developed ethically, taking animals' interests into account and contributing to a positive human-animal relationship? Ethical considerations in the current field of AI ethics and animals include the following. Species bias in algorithms can reinforce the popular image of animals as commodities and edible products (Hagendorff et al, 2023). Datafication in animal farming may lead to the further objectification of animals, leading us to conceive of animals as a 'batch of data' (Bos et al. 2018). Precision livestock farming (PLF) may lead to alienation between farmers and animals and erode farmers' experiential knowledge. It may also create health and welfare risks in case of malfunction or breakdown.

In this assignment topic, you will develop a design for a tool that **manages animal welfare**.

You will need to select a destination context such as agriculture, wild animal management, or laboratory research on animals.

This is not a technical assignment. Hence the grade is not based on technical aspects of the design. You may explore the landscape of commercial, off-the-shelf tools, although you are not required to use any of them. You will share potential tools and strategies with other students.

Client: Mark Ryan, Wageningen University

Primary Student Coach: Manos Alatserakis

References:

Bos, J.M., Bovenkerk, B., Feindt, P.H., and van Dam, Y.K. (2018). The quantified animal: Precision livestock farming and the ethical implications of objectification. *Food Ethics* 2, 77-92.

<https://doi.org/10.1007/s41055-018-00029-x>

Hagendorff, T., Bossert, L.N., Tse, Y.F. and Singer, P. (2023). Speciesist bias in AI: How AI applications perpetuate discrimination and unfair outcomes against animals. *AI and Ethics* 3, 717-734.

<https://doi.org/10.1007/s43681-022-00199-9>

Owe, A. and Baum, S.D. (2021). Moral consideration of nonhumans in the ethics of artificial intelligence. *AI and Ethics* 1, 517–528. <https://doi.org/10.1007/s43681-021-00065-0>

Ryan, M. (2023). The social and ethical impacts of artificial intelligence in agriculture: mapping the agricultural AI literature. *AI & Society* 38, 2473-2485.

Topic EADS

In many countries, the number of young people with an eating disorder has grown in recent years. Yet the waiting time for appropriate treatment is long, leaving some of these young people in a care gap. Recently, experimental interventions have been developed involving Featback, a fully automated, easily accessible system for people who may have eating disorders. Such systems have shown promise in experimental trials. You can find more information at <https://featback.nl/>

In this topic, students explore what role AI might play in an automated system like Featback, and what ethical issues might arise in the process, such as risks to users, misuse, bias, etc. Students are invited to create a mock-up or demo of a user-interface in which AI is integrated. The client is willing to discuss sharing data from an earlier scientific study.

This is not a technical assignment. Hence the grade is not based on technical aspects of the design. You may explore the landscape of commercial, off-the-shelf tools, although you are not required to use any of them. You will share potential tools and strategies with other students.

Client: Alexandra Dingemans, a.dingemans@rivierduinen.nl

Primary Student Coach: Rithic Chellaram Hariharan

Reference:

Rohrbach, P.J., Dingemans, A.E., Spinhoven, P., Van Ginkel, J.R., Fokkema, M., Wilderjans, T.F., Bauer, S. and Van Furth, E.F., 2022. Effectiveness of an online self-help program, expert-patient support, and their combination for eating disorders: Results from a randomized controlled trial. *International Journal of Eating Disorders*, 55(10), pp.1361-1373.

Topic ENSU

AI for sustainability is defined by Hasselbalch & van Wynsberghe (2025) as the use of AI tools to increase sustainability, for example by predicting environmental impacts, optimizing resource use and streamlining production processes. Here at the TU/e, sustainability is identified as a core institutional value, and artificial intelligence is also promoted at many levels. However, the TU/e's "Sustainability Vision" does not mention artificial intelligence.

In this challenge topic, students will draft a new Sustainability Tool for TU/e concerning Research, Education or Operation. It can be based on the existing sustainability self-assessment tool, but it should incorporate AI for sustainability in a creative, well-supported way. It should also be consistent with TU/e sustainability policy.

The existing tool can be found at <https://www.tue.nl/en/our-university/about-the-university/sustainability/research/self-assessment-tool>

This is not a technical assignment. Hence the grade is not based on technical aspects of the design. You may explore the landscape of commercial, off-the-shelf tools, although you are not required to use any of them. You will share potential tools and strategies with other students.

Clients: Philip Nickel & Tijn Borghuis, Eindhoven University of Technology

Primary **Student Coach:** Qin Tao (Tata) Zeng

References:

Berthelot, A., Caron, E., Jay, M., and Lefèvre, L. 2025. Understanding the Environmental Impact of Generative AI Services. *Communications of the ACM* 68, 7: 46–53.

<https://doi.org/10.1145/3725984>

Hasselbalch, G., & van Wynsberghe, A. (2025). Artificial Intelligence, Power, and Sustainability. In Smuha, P., ed. *The Cambridge Handbook of the Law, Ethics and Policy of Artificial Intelligence*. Cambridge University Press. Pp. 117-130.